**Sheet 4** Thu. 15 Oct. 2026 Problem solving: vector, coordinate or geometric method

**1.** Warm-up · Flash [GEO] Complete: $ABCD$ is a parallelogram $\Leftrightarrow \overrightarrow{AB} = \ldots$

**Ans.** $\overrightarrow{AB} = \overrightarrow{DC}$ (or $\overrightarrow{AD} = \overrightarrow{BC}$).

The vertices are named in order: $[AC]$ *et* $[BD]$ are the diagonals.

**2.** Core · Method [REP] $A(0;0)$, $B(4;1)$, $C(5;4)$. Determine $D$ so that $ABCD$ is a parallelogram.

**Ans.** $\overrightarrow{AB} = \overrightarrow{DC}$ : $(4;1) = (5 - x_D; 4 - y_D)$, therefore $D(1;3)$.

**3.** Core + · Write up [DEM] $A(0;0)$, $B(3;1)$, $C(-1;3)$. Prove that $ABC$ is right-angled and isosceles at $A$.

**Ans.** $AB^2 = 10$, $AC^2 = 10$ : $AB = AC$ ; $BC^2 = 16 + 4 = 20 = AB^2 + AC^2$ : converse of Pythagoras' theorem.

Two separate justifications: equal lengths and a right angle.

**4.** Back · Flash [NUM] Write 450,000 in scientific notation.

**Ans.** $4{,}5 \times 10^5$.

**5.** Running thread · Flash [SOL] How many faces, vertices and edges does a cube have?

**Ans.** 6 faces, 8 vertices, 12 edges.

## Lines in the plane

**Sheet 5** Mon. 2 Nov. 2026 Direction vector of a line

**1.** Warm-up · Flash [GEO] Give a vector collinear with $\vec{u}\begin{pmatrix}2\\-4\end{pmatrix}$ with simpler integer coordinates.

**Ans.** $\begin{pmatrix}1\\-2\end{pmatrix}$ (divide by 2).

**2.** Core · Method [GEO] Give a direction vector of $(AB)$ for $A(1;2)$ and $B(4;8)$.

**Ans.** $\overrightarrow{AB}\begin{pmatrix}3\\6\end{pmatrix}$, or $\begin{pmatrix}1\\2\end{pmatrix}$ which is collinear to it.

**3.** Core + · MCQ [GEO] A direction vector of the line through $A(0;1)$ and $B(2;-3)$ is:

**A.** $\begin{pmatrix}1\\-2\end{pmatrix}$ **B.** $\begin{pmatrix}1\\-4\end{pmatrix}$ **C.** $\begin{pmatrix}-4\\-2\end{pmatrix}$ **D.** $\begin{pmatrix}2\\4\end{pmatrix}$

**Ans.** A. $\begin{pmatrix}1\\-2\end{pmatrix}$

*Ici* $\overrightarrow{AB}\begin{pmatrix}2\\-4\end{pmatrix}$ ; collinearity is tested with the determinant.

**4.** Back · Flash [STA] Marks: 8, 12, 12, 15, 18. Calculate the mean, the median and the range.

**Ans.** Mean 13; median 12; range 10.

**5.** Running thread · Property [DEM] $ABC$ is right-angled at $A$ and $I$ is the midpoint of $[BC]$. What can be said about $IA$, $IB$ and $IC$ ? Why?

**Ans.** $IA = IB = IC$ : the midpoint of the hypotenuse is the centre of the circumscribed circle.

**Sheet 6** Tue. 3 Nov. 2026 Cartesian equation of a line

**1.** Warm-up · Flash [SUB] Calculate $3x - 2y + 1$ for $x = 1$ and $y = -2$.

**Ans.** $3 + 4 + 1 = 8$.

**2.** Core · Method [GEO] The line $d$ has equation $x + 3y - 6 = 0$. Give a direction vector and a point of $d$.

**Ans.** For $ax + by + c = 0$, $\begin{pmatrix}-b\\a\end{pmatrix}$ directs the line: $\begin{pmatrix}-3\\2\end{pmatrix}$. The point $(3;0)$ works: $6 + 0 - 6 = 0$.

**3.** Core + · True/false [GEO] True or false? Justify. The lines $x + 3y - 6 = 0$ and $4x + 6y + 1 = 0$ are parallel.

**Ans.** True. Direction vectors $\begin{pmatrix}-3\\2\end{pmatrix}$ and $\begin{pmatrix}-6\\4\end{pmatrix}$ collinear; they are distinct: $(3;0)$ does not belong to the second line ($12 + 1 \neq 0$).

**4.** Back · Table [LEC] Mode of transport of 30 pupils.

| Transport | Bus | Bicycle | Walking | Car |
|---|---|---|---|---|
| Count | 12 | 5 | 8 | 5 |

Give the share of pupils coming by bus and the angle of the "bicycle" sector in a pie chart.

**Ans.** $\frac{12}{30} = 40\,\%$ ; $\frac{5}{30} \times 360° = 60°$.

**5.** Running thread · Flash [REP] On a number line, $A$ has x-coordinate $-2$ and $B$ with x-coordinate 3. Give $AB$ and the x-coordinate of the midpoint of $[AB]$.

**Ans.** $AB = 3 - (-2) = 5$ ; midpoint: $\frac{-2+3}{2} = 0{,}5$.

**Sheet 7** Thu. 5 Nov. 2026 Reduced equation and slope of a non-vertical line

**1.** Warm-up · Flash [EQ] Isolate $y$ in $x + y - 5 = 0$.

**Ans.** $y = -2x + 5$.

**2.** Core · Method [GEO] Give the slope and y-intercept of the line $x - 2y + 4 = 0$.

**Ans.** $-2y = -3x - 4$, $y = \frac{3}{2}x + 2$ : slope $\frac{3}{2}$, y-intercept 2.

Pitfall: dividing by $-2$ each term (signs).

**3.** Core + · MCQ [GEO] The slope of the line $(AB)$ with $A(1;2)$ and $B(5;0)$ is:

**A.** 2 **B.** $\frac{1}{2}$ **C.** $-2$ **D.** $-\frac{1}{2}$

**Ans.** D. $-\frac{1}{2}$

Slope $= \frac{y_B - y_A}{x_B - x_A} = \frac{-2}{4}$.

**4.** Back · Flash [CAL] Calculate $\frac{3}{4} + \frac{5}{6}$.

**Ans.** $\frac{9}{12} + \frac{10}{12} = \frac{19}{12}$.

**5.** Running thread · Problem [DIV] Two gears have 12 and 18 teeth. After how many turns of each do they return together to their starting position?

**Ans.** LCM $(12\,;\,18) = 36$ teeth: 3 turns of the first, 2 turns of the second.

**Sheet 8** Mon. 9 Nov. 2026 Draw a line given by an equation

**1.** Warm-up · Flash [IMG] For $y = -2x + 3$, calculate $y$ when $x = 0$, then when $x = 2$.

**Ans.** $y = 3$ then $y = -1$.

**2.** Core · Method [REG] Give two points for drawing the line $x + y - 4 = 0$.

**Ans.** For $x = 0$ : $y = 4$; for $x = 2$ : $y = 0$. Points $(0;4)$ and $(2;0)$.

**3.** Core + · Flash [GEO] What is the line with equation $x = 3$ ? Give a direction vector. Does it have a slope?

**Ans.** The vertical line through $(3;0)$ ; $\begin{pmatrix}0\\1\end{pmatrix}$ directs it; it has no slope; Cartesian equation $x - 3 = 0$.

Cannot be written $y = mx + p$.

**4.** Back · Flash [CMP] Compare $\frac{3}{7}$ and $\frac{4}{9}$

**Ans.** Cross-multiplication: $3 \times 9 = 27 < 28 = 4 \times 7$, so $\frac{3}{7} < \frac{4}{9}$.

**5.** Running thread · Flash [EQ] Solve in $\mathbb{R}$ : $5x - 7 = 2x + 8$.

**Ans.** $x = 5$

Pitfall: gathering the x terms on one side while changing signs correctly.

**Sheet 9** Tue. 10 Nov. 2026 Determine an equation of a line: two points, point and direction vector, point and slope

**1.** Warm-up · Flash [GEO] Slope of the line through $(1;3)$ and $(3;7)$ ?

**Ans.** $\dfrac{7-3}{3-1}=2$.

**2.** Core · Method [GEO] Reduced equation of the line with slope 2 passing through $A(1;3)$.

**Ans.** $y-3=2(x-1)$, i.e. $y=2x+1$.

Check: $2\times 1+1=3$.

**3.** Core + · Method [GEO] Line through $A(1;2)$ with direction vector $\vec{u}\begin{pmatrix}2\\-1\end{pmatrix}$ : Cartesian equation by the determinant of $\overrightarrow{AM}$ and $\vec{u}$.

**Ans.** $M(x;y)\in d \Leftrightarrow \det\left(\overrightarrow{AM},\vec{u}\right)=0 : (x-1)(-1)-(y-2)\times 2=0$, i.e. $x+2y-5=0$.

Check: $1+4-5=0$ ; $\begin{pmatrix}-2\\1\end{pmatrix}$ is indeed collinear with $\begin{pmatrix}2\\-1\end{pmatrix}$.

**4.** Back · Flash [PUI] Simplify $2^{5}\times 2^{-3}$.

**Ans.** $2^{5-3}=2^2=4$.

**5.** Running thread · Table → formula [REG] An affine function $f$ satisfies: $f(0)=-1, f(1)=1, f(2)=3$. Give $f(x)$.

**Ans.** Slope 2, y-intercept $-1$ : $f(x)=2x-1$.

**Sheet 10** Thu. 12 Nov. 2026 Lines and affine functions: linking the representations

**1.** Warm-up · Flash [IMG] $f(x)=-3x+4$. Calculate $f(2)$ and the preimage of 10.

**Ans.** $f(2)=-2$ ; $-3x+4=10 \Leftrightarrow x=-2$.

**2.** Core · Method [REG] Determine the affine function $f$ such that $f(1)=1$ and $f(3)=5$.

**Ans.** Slope $\dfrac{5-1}{3-1}=2$ ; $f(x)=2x+p$ with $2+p=1 : f(x)=2x-1$.

**3.** Core + · True/false [REG] True or false? Justify. Every line in the plane is the graph of an affine function.

**Ans.** False. Counterexample: the vertical line $x=3$ is not the graph of any function (the number 3 would have infinitely many images).

**4.** Back · Flash [PRB] A fair 6-sided die is rolled. Probability of getting a multiple of 3?

**Ans.** Outcomes 3 and 6: $\dfrac{2}{6}=\dfrac{1}{3}$.

**5.** Running thread · Flash [GEO] $ABC$ is right-angled at $A$ with $AB=6$ cm and $AC=8$ cm. Calculate $BC$.

**Ans.** $BC^2=36+64=100$, so $BC=10$ cm.

**Sheet 11** Mon. 16 Nov. 2026 Collinearity, parallelism, intersection of lines

**1.** Warm-up · Flash [SUB] The point $(3;5)$ lie on the line $y=2x-1$ ?

**Ans.** $2\times 3-1=5$ : Yes.

**2.** Core · Method [GEO] Relative positions of $d : y=3x+1$ with $d' : y=3x-2$, then with $d'' : y=-x+1$.

**Ans.** $d$ and $d'$ : same slope, different y-intercepts: strictly parallel; $d$ and $d''$ : slopes different: intersecting.

**3.** Core + · Method [GEO] Show that $A(1;4), B(2;7)$ and $C(4;13)$ are collinear in two ways.

**Ans.** (i) All satisfy $y=3x+1$. (ii) $\overrightarrow{AB}\begin{pmatrix}1\\3\end{pmatrix}, \overrightarrow{AC}\begin{pmatrix}3\\9\end{pmatrix}$ : $\det\left(\overrightarrow{AB},\overrightarrow{AC}\right)=9-9=0$.

**4.** Back · Flash [PRO] On a map with scale $\dfrac{1}{50\,000}$, two towns are 4 cm apart. What is the actual distance?

**Ans.** $4\times 50\,000=200\,000$ cm $=2$ km.